Predicting Water Well Functionality in Tanzania

A Classification Model for the Tanzanian Ministry of Water

59,400	26	3
Wells in Dataset	Features Used	Classes

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- *Phase 3 Project*
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The Problem

60M+ People

Tanzania's population depends on waterpoints as their primary source of clean water.

38% Non-Functional

More than a third of wells are broken or at risk — leaving communities without safe water.

Reactive Maintenance

Repairs are only made after wells fail. There is no system to predict which wells need attention.

Our Solution

A multi-class classifier that predicts well status from available features — enabling the Ministry to act before wells fail.

Functional

Well is working.
No action needed.

Functional Needs Repair

Well is at risk.
Schedule maintenance.

Non Functional

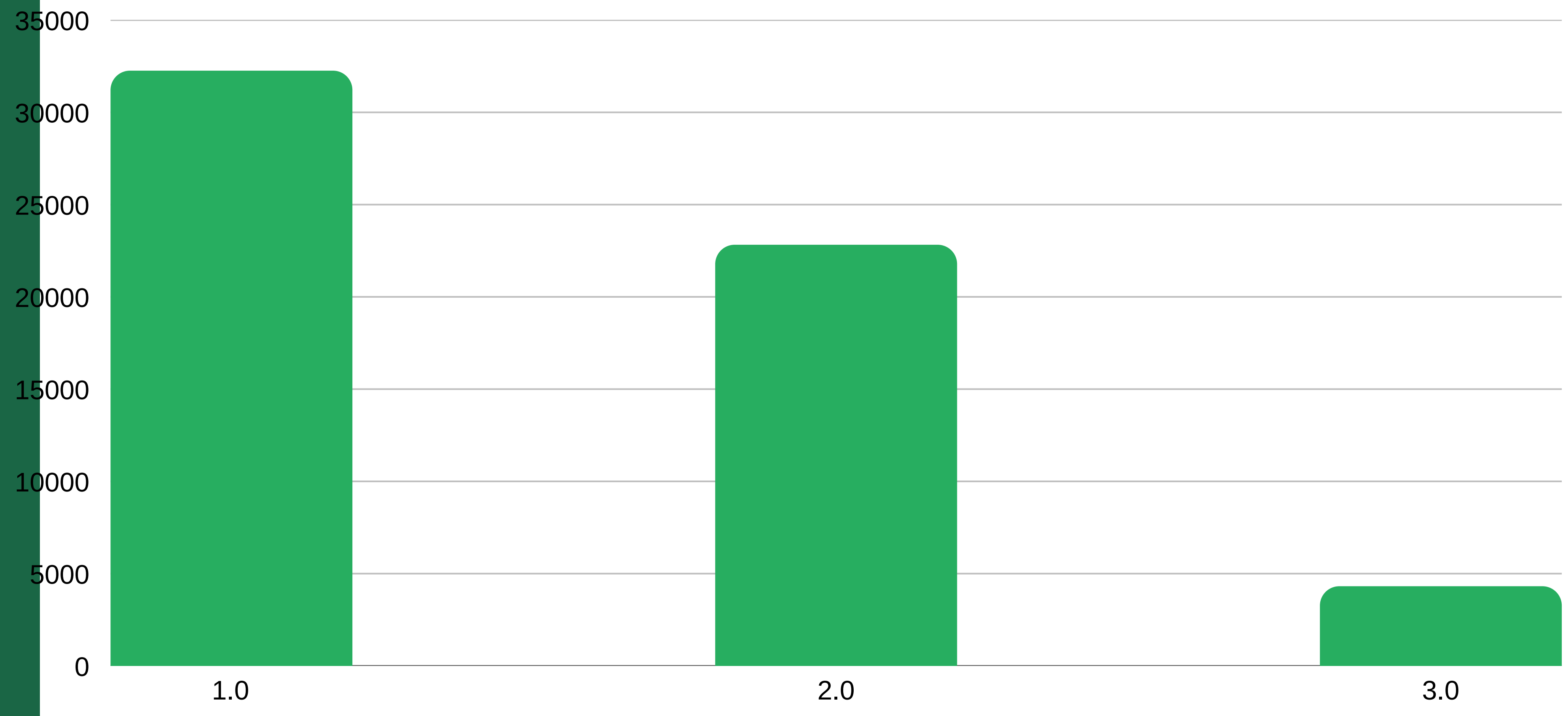
Well is broken.
Prioritise urgent repair.

The Data

Source: DrivenData — Pump It Up Competition | Tanzanian Ministry of Water

59,400	40	26	60/20/20
Water wells	Raw features	Features used	Train/Val/Test split

Class Distribution



Key Predictive Features

- 📍 GPS coordinates & altitude
- 💧 Water quantity & quality
- ⚙️ Extraction type & pump type
- 📅 Construction year
- 🏛️ Funder & installer
- 👥 Population around the well

Modeling Approach

We built models iteratively — each justified by the limitations of the one before it.

1

Dummy Classifier

Baseline — always predicts most frequent class (~54% accuracy). Sets the performance floor.

→

2

Logistic Regression

Linear model — tests whether classes are linearly separable. Requires feature scaling.

→

3

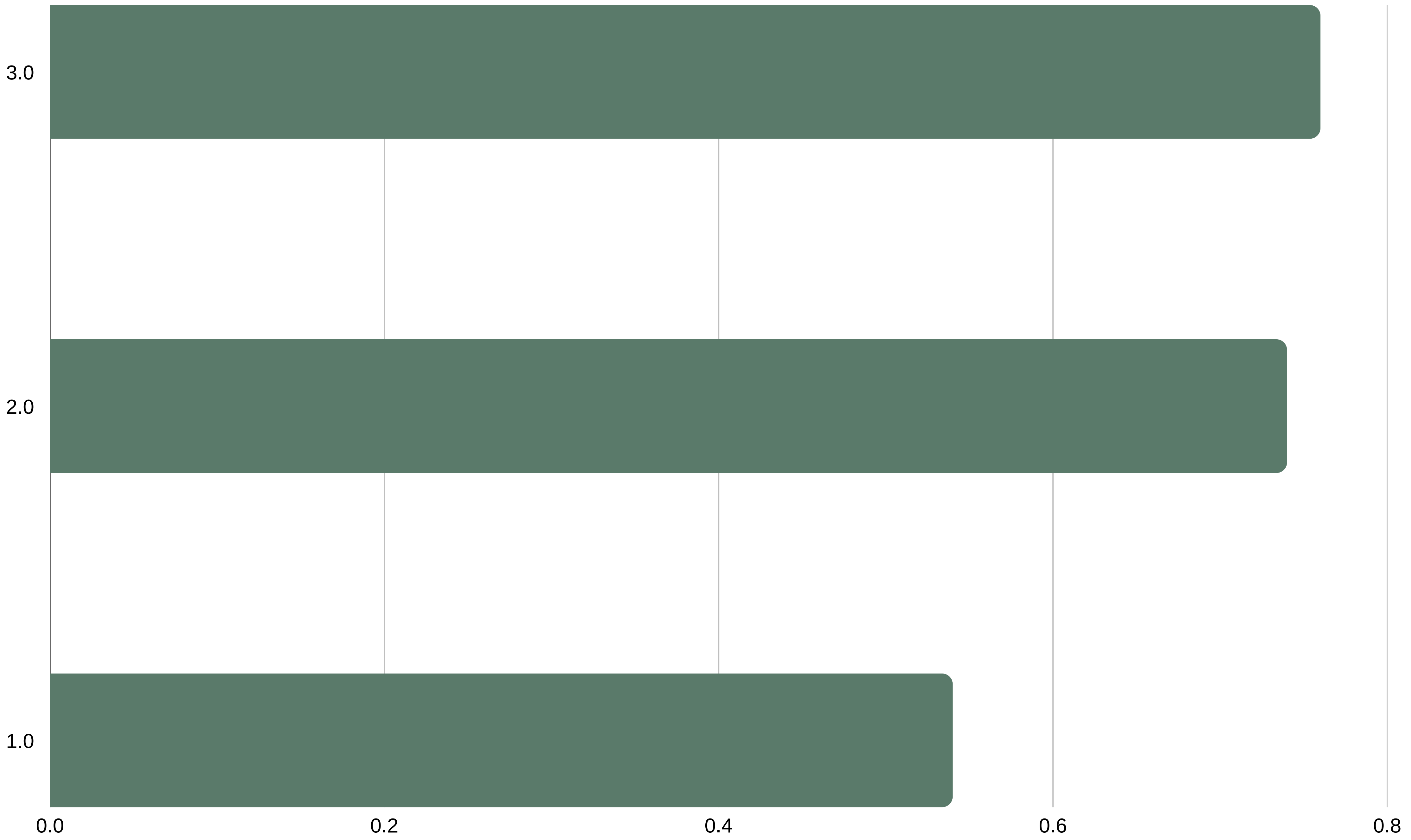
Decision Tree ★

Captures non-linear patterns. Interpretable, no scaling needed. Best performing model.

Model Results

Our final model — the Decision Tree — was chosen for its ability to capture non-linear patterns and provide interpretable predictions for stakeholders.

Validation Accuracy by Model



~76%

Decision Tree Accuracy

~74%

Logistic Reg. Accuracy

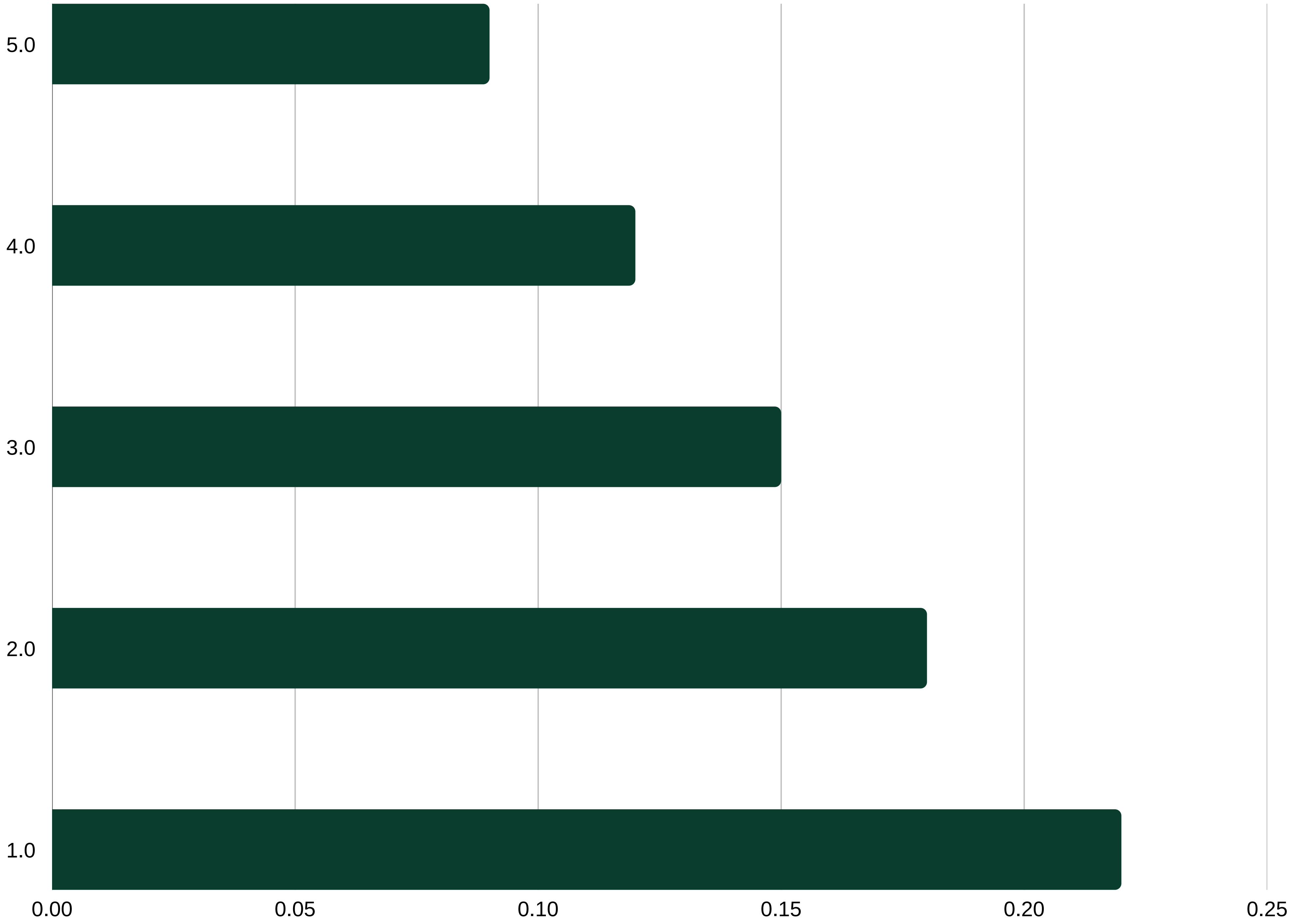
~54%

Dummy Baseline Accuracy

Key Findings

What the Decision Tree revealed about water well functionality in Tanzania.

Most Important Features



Water Quantity is Key

Wells with 'dry' or 'insufficient' quantity are far more likely to be non-functional.

Older Wells Fail More

Construction year is a strong predictor — older wells have significantly higher failure rates.

Minority Class is Hard

'Functional needs repair' (7% of data) is the hardest class to predict due to class imbalance.

Recommendations

Deploy for Triage

Use model predictions to flag non-functional wells and prioritize repair crew dispatch.

Track High-Risk Regions

Focus inspection budgets on regions where the model predicts clusters of failing wells.

NGO Coordination

Share predictions with UNICEF and Danida to coordinate targeted preventive maintenance.

Retrain Annually

Well status changes over time — refresh the model with updated survey data each year.

From Reactive to Proactive.

With a reliable predictive model, the Tanzanian Ministry of Water can move from reacting to broken wells to preventing them — ensuring more communities have consistent access to clean water.

THANK YOU !!

QUESTIONS?????